Date

Read Slides too
↑ Lec#17 (GCR)

LECTURE-14

Monday
19/10/26.

☐ Instruction Level Parallelism:

RISC-V	CISC	VLIW: High parallel
→ Simple hardware	• Complex Ins. set computers	• Heavy compiler is adv.
• [redacted] simple	• [redacted] hardware ↑ complex.	but now heavily dependent
• Pipeline Simple	• Complex Pipelining	on compiler too
• ISA	• Supports Intel x86	• Combined Instruc. here
• fixed length Instruc.	• variable length	& multiple Instruc. runs
• One instruction	• large instruc. set	• Hardware simple
do one task	do multiple tasks	• All instruction independent
• RISC-V, ARM.	• faster but per instruc.	& it makes a word
	time increases	to executed it
↓ ld x1, 0(x2)	add + memory + Store	• High cost of compiler
add x1, x1, 100	• Eg: x86	∴ discontinued. (Intel + HP)
sd, x1, 0(x2)		

★ CISC have multiple operation in single Instru. VLIW have multiple instructions so a lot of operations

VLIW: Very long Instruction set
High compiler & Hardware Complexity
E.g.: Itanium

⇒ Super Scalar: Even more parallelism but using "hardware" support instead of "Compiler".

Regd No.

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Q: Base CPI = 1 Avg. No. of cycles per Instruction
Total Instructions = 100
data Hazard stalls = 10 cycles
Branch " " = 20 cycles
Structural " " = 15 cycles

$$CPI = \frac{\text{Total Cycles}}{\text{Total Instructions}}$$

1 CPI per Instruction

- Cycles for 100 instruction = 1 x 100
  = 100
- Other cycles are of Hazard stalls = 10 + 20 + 15
  = 45

$$CPI = \frac{100 + 45}{100} = 1.45$$

Total no. of Instruc.

CamScanner